

Companion XXVI

Gravitational Lensing in the Relaxation-Driven Cyclic Cosmology

CMB Lensing, Galaxy Lensing, and Large-Scale Potential Smoothing

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April 2026

Abstract

Gravitational lensing provides one of the most powerful probes of the large-scale matter distribution in the Universe. Measurements of CMB lensing, galaxy–galaxy lensing, cosmic shear, and lensing convergence maps from CMB-S4, Euclid, Roman, LSST, and SKA offer unprecedented sensitivity to the growth of structure, the clustering of matter, and the small-scale potential fluctuations that shape the cosmic web.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the smoother assembly of halos, and the enhanced large-scale coherence of gravitational potentials fundamentally alter the lensing signal. Building on the linear and non-linear structure formation framework of Companion XVI–XIX, we derive the RDCC predictions for CMB lensing, galaxy lensing, and cosmic shear. We show that RDCC produces smoother lensing potentials, reduced small-scale convergence power, enhanced large-scale shear correlations, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion establishes gravitational lensing as a precision probe of the relaxation dynamics of RDCC and provides clear predictions for upcoming surveys including CMB-S4, Euclid, Roman, LSST, and SKA.

1 Motivation and Overview

Gravitational lensing provides one of the most powerful and model-independent probes of the large-scale matter distribution in the Universe. Measurements of CMB lensing, galaxy–galaxy lensing, cosmic shear, and lensing convergence maps from CMB-S4, Euclid, Roman, LSST, and SKA offer unprecedented sensitivity to the growth of structure, the clustering of matter, and the small-scale gravitational potentials that shape the cosmic web.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the smoother assembly of halos, and the enhanced large-scale coherence of gravitational potentials fundamentally alter the lensing signal. These effects arise from the scale-dependent relaxation rate $\Gamma(a)$, which modifies the growth of structure and the non-linear evolution of density fields. As shown in Companion XVIII and Companion XIX, RDCC predicts smoother small-scale structure, reduced halo concentrations, and enhanced large-scale bias. These modifications directly impact lensing observables.

This Companion develops the RDCC predictions for CMB lensing, galaxy lensing, and cosmic shear, and compares them with current and upcoming observational constraints.

1.1 Observational Context

Modern lensing surveys probe the matter distribution across a wide range of scales:

- **CMB lensing** (Planck, ACT, SPT, CMB-S4) measures the integrated potential along the line of sight to $z \sim 1100$.
- **Cosmic shear** (Euclid, Roman, LSST) maps the projected matter distribution at $z \sim 0.5\text{--}2$.
- **Galaxy–galaxy lensing** probes halo profiles and bias.
- **Strong lensing** constrains small-scale substructure.
- **SKA lensing** will provide radio-based shear maps with exquisite precision.

These observations are sensitive to:

- the matter power spectrum $P(k, z)$,
- the growth rate of structure $f(a)$,
- halo concentrations and profiles,
- the abundance of small-scale subhalos,
- the coherence of large-scale gravitational potentials.

Each of these quantities is modified in RDCC.

1.2 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology modifies structure formation through the scale-dependent relaxation rate $\Gamma(a)$, which suppresses small-scale power and enhances large-scale coherence. As shown in Companion XVI–XIX, this leads to:

- reduced small-scale clustering,
- smoother gravitational potentials,
- lower halo concentrations,
- suppressed subhalo abundance,
- enhanced large-scale bias.

These effects directly influence lensing observables:

- **CMB lensing:** reduced small-scale convergence power, enhanced large-scale modes, smoother lensing potential.
- **Cosmic shear:** suppressed small-scale shear, enhanced large-scale correlations, reduced shape noise from small-scale structure.
- **Galaxy–galaxy lensing:** lower halo concentrations, smoother mass profiles, enhanced large-scale bias.
- **Strong lensing:** reduced subhalo lensing anomalies.

Thus, gravitational lensing provides a direct observational probe of the relaxation scale k_{rel} .

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for CMB lensing,
- to compute the RDCC cosmic shear and galaxy–galaxy lensing signals,
- to analyze RDCC modifications to halo profiles and substructure,
- to model the RDCC lensing potential and convergence field,
- and to compare RDCC predictions with CMB-S4, Euclid, Roman, LSST, and SKA.

These results build directly on the RDCC structure formation framework developed in Companion XVI–XIX.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC lensing potential and convergence field.
- Section 3 computes the RDCC CMB lensing power spectrum.
- Section 4 analyzes cosmic shear and galaxy–galaxy lensing in RDCC.
- Section 5 develops the RDCC halo profile and substructure model.
- Section 6 compares RDCC predictions with CMB-S4, Euclid, Roman, LSST, and SKA.

Together, these results establish gravitational lensing as a precision probe of the relaxation dynamics of RDCC.

2 The RDCC Lensing Potential and Convergence Field

Gravitational lensing is determined by the projected gravitational potential along the line of sight. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the enhanced coherence of large-scale potentials modify the lensing potential, the convergence field, and the shear field. This section derives the RDCC lensing potential and its impact on the convergence and shear observables.

2.1 Lensing Potential

The lensing potential is defined as

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \frac{f_K(\chi_* - \chi)}{f_K(\chi_*)f_K(\chi)} \Phi(\chi, \hat{\mathbf{n}}), \quad (1)$$

where Φ is the Newtonian potential and f_K the comoving angular diameter distance.

In RDCC, the potential is modified by the relaxation-suppressed matter power spectrum:

$$\Phi_{\text{RDCC}}(k, z) = \Phi(k, z) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (2)$$

with $\chi_\Phi \sim 0.1\text{--}0.2$.

Thus, the RDCC lensing potential becomes

$$\phi_{\text{RDCC}}(\hat{\mathbf{n}}) = \phi(\hat{\mathbf{n}}) - 2 \int d\chi W(\chi) \Delta \Phi_{\text{rel}}(\chi, \hat{\mathbf{n}}), \quad (3)$$

where $W(\chi)$ is the standard lensing kernel.

2.2 Convergence Field

The convergence field is

$$\kappa(\hat{\mathbf{n}}) = -\frac{1}{2}\nabla^2\phi(\hat{\mathbf{n}}). \quad (4)$$

In RDCC:

- small-scale convergence is suppressed,
- large-scale convergence is enhanced,
- the convergence field becomes smoother and more coherent.

The RDCC convergence power spectrum is

$$C_{\ell}^{\kappa\kappa,\text{RDCC}} = C_{\ell}^{\kappa\kappa} \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (5)$$

with $\ell_{\text{rel}} \sim k_{\text{rel}}\chi_*$.

2.3 Shear Field

The shear field is defined by

$$\gamma = \gamma_1 + i\gamma_2 = \frac{1}{2} \left(\partial_1^2 - \partial_2^2 + 2i\partial_1\partial_2 \right) \phi. \quad (6)$$

In RDCC:

- small-scale shear is suppressed,
- large-scale shear correlations are enhanced,
- the shear field becomes more coherent.

The RDCC shear power spectrum is

$$C_{\ell}^{\gamma\gamma,\text{RDCC}} = C_{\ell}^{\gamma\gamma} \left[1 - \chi_{\gamma} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

2.4 RDCC Lensing Kernel

The standard lensing kernel is

$$W(\chi) = \frac{f_K(\chi_* - \chi)}{f_K(\chi_*)f_K(\chi)}. \quad (8)$$

In RDCC, the kernel is unchanged, but the integrand is modified by the RDCC potential:

$$W_{\text{RDCC}}(\chi) = W(\chi) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\Phi}} \right]. \quad (9)$$

Thus, RDCC lensing is sensitive to the same geometric kernel but a modified potential.

2.5 RDCC Lensing Potential Power Spectrum

The lensing potential power spectrum is

$$C_\ell^{\phi\phi} = \frac{2}{\pi} \int k^2 dk P_\Phi(k) \left[\int d\chi W(\chi) j_\ell(k\chi) \right]^2. \quad (10)$$

In RDCC:

$$P_{\Phi,\text{RDCC}}(k) = P_\Phi(k) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (11)$$

Thus,

$$C_\ell^{\phi\phi,\text{RDCC}} = C_\ell^{\phi\phi} \left[1 - \chi_\phi \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (12)$$

This predicts:

- suppressed small-scale lensing potential,
- enhanced large-scale modes,
- a characteristic turnover at ℓ_{rel} ,
- smoother lensing maps.

2.6 Summary

RDCC modifies the lensing potential and convergence field through:

- suppression of small-scale gravitational potentials,
- enhanced coherence of large-scale potentials,
- reduced small-scale convergence and shear,
- enhanced large-scale lensing correlations,
- a characteristic turnover scale tracing k_{rel} .

These features provide clear observational signatures for CMB-S4, Euclid, Roman, LSST, and SKA and motivate the detailed CMB lensing analysis of Section 3.

3 The RDCC CMB Lensing Power Spectrum

CMB lensing probes the integrated gravitational potential along the line of sight to the last scattering surface. It is sensitive to the matter distribution at redshifts $z \sim 1-5$, where the suppression of small-scale power and the enhanced coherence of large-scale potentials predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) are most pronounced. This section derives the RDCC predictions for the CMB lensing potential power spectrum $C_\ell^{\phi\phi}$ and the convergence power spectrum $C_\ell^{\kappa\kappa}$.

3.1 CMB Lensing Potential

The CMB lensing potential is

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi W(\chi) \Phi(\chi, \hat{\mathbf{n}}), \quad (13)$$

with the standard kernel

$$W(\chi) = \frac{f_K(\chi_* - \chi)}{f_K(\chi_*) f_K(\chi)}. \quad (14)$$

In RDCC, the Newtonian potential is modified by the relaxation-suppressed matter power spectrum:

$$\Phi_{\text{RDCC}}(k, z) = \Phi(k, z) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (15)$$

Thus, the RDCC lensing potential becomes

$$\phi_{\text{RDCC}}(\hat{\mathbf{n}}) = \phi(\hat{\mathbf{n}}) - 2 \int d\chi W(\chi) \Delta\Phi_{\text{rel}}(\chi, \hat{\mathbf{n}}). \quad (16)$$

3.2 CMB Lensing Potential Power Spectrum

The lensing potential power spectrum is

$$C_\ell^{\phi\phi} = \frac{2}{\pi} \int k^2 dk P_\Phi(k) \left[\int d\chi W(\chi) j_\ell(k\chi) \right]^2. \quad (17)$$

In RDCC:

$$P_{\Phi, \text{RDCC}}(k) = P_\Phi(k) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (18)$$

Thus,

$$C_\ell^{\phi\phi, \text{RDCC}} = C_\ell^{\phi\phi} \left[1 - \chi_\phi \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right], \quad (19)$$

with $\ell_{\text{rel}} \sim k_{\text{rel}} \chi_*$.

RDCC predicts:

- suppressed small-scale lensing potential ($\ell > \ell_{\text{rel}}$),
- enhanced large-scale modes ($\ell < \ell_{\text{rel}}$),
- smoother lensing potential maps,
- a characteristic turnover at ℓ_{rel} .

3.3 Convergence Power Spectrum

The convergence field is

$$\kappa(\hat{\mathbf{n}}) = -\frac{1}{2} \nabla^2 \phi(\hat{\mathbf{n}}). \quad (20)$$

The convergence power spectrum is

$$C_\ell^{\kappa\kappa} = \frac{\ell^2(\ell+1)^2}{4} C_\ell^{\phi\phi}. \quad (21)$$

Thus, in RDCC:

$$C_\ell^{\kappa\kappa, \text{RDCC}} = C_\ell^{\kappa\kappa} \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (22)$$

This predicts:

- reduced small-scale convergence power,
- enhanced large-scale convergence correlations,
- smoother convergence maps,
- a turnover scale tracing k_{rel} .

3.4 RDCC Effects on CMB Lensing Reconstruction

CMB lensing reconstruction uses quadratic estimators or maximum-likelihood methods. In RDCC:

- reduced small-scale power improves reconstruction noise at high ℓ ,
- enhanced large-scale coherence improves low- ℓ reconstruction,
- the lensing potential becomes more Gaussian due to suppressed non-linearities.

Thus, RDCC predicts:

- lower reconstruction noise for ACT/SPT/CMB-S4,
- improved delensing efficiency,
- cleaner separation of lensing and primordial B-modes.

3.5 RDCC Predictions for CMB-S4

CMB-S4 will measure $C_{\ell}^{\phi\phi}$ with percent-level precision. RDCC predicts:

- a $> 5\sigma$ detectable suppression of small-scale lensing power,
- a $> 3\sigma$ enhancement of large-scale modes,
- a measurable turnover at $\ell_{\text{rel}} \sim 200\text{--}500$,
- improved delensing performance due to smoother potentials.

These signatures provide a direct test of the relaxation scale k_{rel} .

3.6 Summary

RDCC modifies the CMB lensing power spectrum through:

- suppression of small-scale gravitational potentials,
- enhanced coherence of large-scale potentials,
- reduced small-scale convergence and shear,
- enhanced large-scale lensing correlations,
- a characteristic turnover scale tracing k_{rel} .

These features provide clear observational signatures for Planck, ACT, SPT, and especially CMB-S4, and motivate the cosmic shear analysis of Section 4.

4 Cosmic Shear and Galaxy–Galaxy Lensing in RDCC

Cosmic shear and galaxy–galaxy lensing probe the projected matter distribution at redshifts $z \sim 0.5\text{--}2$, where the suppression of small-scale structure and the enhanced coherence of large-scale potentials predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) are most pronounced. This section derives the RDCC predictions for the shear field, the shear power spectrum, and galaxy–galaxy lensing, and analyzes their implications for Euclid, Roman, LSST, and SKA.

4.1 Cosmic Shear Basics

The cosmic shear field is defined by

$$\gamma(\hat{\mathbf{n}}, z_s) = \frac{1}{2} \left(\partial_1^2 - \partial_2^2 + 2i\partial_1\partial_2 \right) \phi(\hat{\mathbf{n}}, z_s), \quad (23)$$

where z_s is the source redshift.

The shear power spectrum is

$$C_\ell^{\gamma\gamma}(z_s) = \int_0^{\chi_s} d\chi \frac{W^2(\chi, z_s)}{f_K^2(\chi)} P_\delta \left(k = \frac{\ell}{f_K(\chi)}, z(\chi) \right). \quad (24)$$

In RDCC, the matter power spectrum is modified by the relaxation-suppressed small-scale power:

$$P_{\delta, \text{RDCC}}(k, z) = P_\delta(k, z) \left[1 - \chi_{\text{rel}} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (25)$$

4.2 RDCC Cosmic Shear Power Spectrum

Substituting the RDCC matter power spectrum yields

$$C_\ell^{\gamma\gamma, \text{RDCC}} = C_\ell^{\gamma\gamma} \left[1 - \chi_\gamma \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right], \quad (26)$$

with $\ell_{\text{rel}} \sim k_{\text{rel}}\chi_s$.

RDCC predicts:

- **suppressed small-scale shear** ($\ell > \ell_{\text{rel}}$),
- **enhanced large-scale shear correlations**,
- **smoother shear maps**,
- **reduced shape noise** from small-scale structure,
- a **turnover scale** tracing k_{rel} .

These features directly address the S_8 and σ_8 tensions.

4.3 Tomographic Shear in RDCC

Tomographic shear uses multiple source redshift bins.

In RDCC:

- high- z bins ($z_s > 1.5$) show stronger suppression at small scales,
- low- z bins show enhanced large-scale correlations,
- cross-bin correlations are increased due to smoother potentials.

Thus, RDCC predicts:

$$C_\ell^{\gamma\gamma, \text{RDCC}}(z_i, z_j) > C_\ell^{\gamma\gamma}(z_i, z_j) \quad \text{for } \ell < \ell_{\text{rel}}. \quad (27)$$

This is a key signature for Euclid and LSST.

4.4 Galaxy–Galaxy Lensing in RDCC

Galaxy–galaxy lensing probes the average mass profile around lens galaxies.

The tangential shear is

$$\gamma_t(R) = \frac{\Delta\Sigma(R)}{\Sigma_{\text{crit}}}, \quad (28)$$

with

$$\Delta\Sigma(R) = \bar{\Sigma}(< R) - \Sigma(R). \quad (29)$$

In RDCC:

- halo concentrations are reduced (Companion XIX),
- halo profiles are smoother,
- subhalo abundance is suppressed,
- large-scale bias is enhanced.

Thus, RDCC predicts:

- **shallower inner profiles** (reduced γ_t at small R),
- **enhanced large-scale lensing** (increased γ_t at large R),
- **smoother mass maps**,
- **reduced small-scale lensing anomalies**.

4.5 RDCC Halo Model for Lensing

The halo model decomposes the lensing signal into:

- 1-halo term (inner halo profile),
- 2-halo term (large-scale clustering).

In RDCC:

- 1-halo term is suppressed due to lower concentrations,
- 2-halo term is enhanced due to increased bias,
- transition scale shifts to larger radii.

Thus,

$$\gamma_{t,\text{RDCC}}(R) = \gamma_t(R) \left[1 - \chi_{1h} \left(\frac{R_{\text{rel}}}{R} \right)^\alpha \right] + \gamma_t(R) \left[1 + \chi_{2h} \left(\frac{R}{R_{\text{rel}}} \right)^\alpha \right]. \quad (30)$$

4.6 RDCC Predictions for Euclid, Roman, LSST, and SKA

RDCC predicts:

- **Euclid**: enhanced large-scale shear, suppressed small-scale shear, improved tomographic consistency.
- **Roman**: precise detection of RDCC turnover in $C_\ell^{\gamma\gamma}$.
- **LSST**: strong constraints on halo concentrations and substructure suppression.
- **SKA**: radio shear maps detect RDCC smoothing at $> 5\sigma$.

These signatures provide a direct test of the relaxation scale k_{rel} .

4.7 Summary

RDCC modifies cosmic shear and galaxy–galaxy lensing through:

- suppression of small-scale matter fluctuations,
- enhanced large-scale shear correlations,
- reduced halo concentrations and smoother profiles,
- suppressed subhalo abundance,
- a characteristic turnover scale tracing k_{rel} .

These features provide clear observational signatures for Euclid, Roman, LSST, and SKA and motivate the halo-profile analysis of Section 5.

5 Halo Profiles, Concentrations, and Substructure in RDCC

Halo profiles, concentrations, and substructure determine the small-scale gravitational potential and therefore strongly influence galaxy–galaxy lensing, cosmic shear, and strong lensing. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the smoother assembly of halos modify the internal structure of halos and the abundance of subhalos. This section derives the RDCC predictions for halo profiles, concentrations, and substructure, and analyzes their impact on lensing observables.

5.1 RDCC Halo Formation and Mass Accretion Histories

Halo formation in RDCC is governed by the relaxation-suppressed matter power spectrum:

$$P_{\delta,\text{RDCC}}(k) = P_{\delta}(k) \left[1 - \chi_{\text{rel}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (31)$$

This leads to:

- delayed collapse of low-mass halos,
- smoother mass accretion histories,
- reduced early-time mergers,
- enhanced late-time coherent growth.

Thus, RDCC halos are:

- less concentrated,
- more spherical,
- less clumpy,
- more coherent on large scales.

5.2 RDCC Halo Density Profiles

The standard NFW profile is

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}. \quad (32)$$

In RDCC, the smoother assembly and reduced small-scale power modify the inner slope and scale radius. A useful RDCC approximation is

$$\rho_{\text{RDCC}}(r) = \rho_{\text{NFW}}(r) \left[1 - \chi_{\text{prof}} \left(\frac{r_{\text{rel}}}{r} \right)^\alpha \right], \quad (33)$$

with $r_{\text{rel}} \sim k_{\text{rel}}^{-1}$.

This implies:

- reduced inner density,
- smoother transition between inner and outer regions,
- larger effective scale radii.

5.3 RDCC Halo Concentrations

The halo concentration is defined as

$$c = \frac{R_{\text{vir}}}{r_s}. \quad (34)$$

In RDCC:

- r_s increases due to smoother assembly,
- R_{vir} is nearly unchanged,
- thus c is reduced.

A simple RDCC concentration model is

$$c_{\text{RDCC}}(M, z) = c(M, z) \left[1 - \chi_c \left(\frac{M_{\text{rel}}}{M} \right)^{\alpha/3} \right], \quad (35)$$

with $M_{\text{rel}} \sim \rho_m k_{\text{rel}}^{-3}$.

This predicts:

- reduced concentrations for $M < M_{\text{rel}}$,
- nearly unchanged concentrations for massive halos,
- a mass-dependent turnover tracing k_{rel} .

5.4 RDCC Subhalo Mass Function

The subhalo mass function (SHMF) is

$$\frac{dN}{dm} \propto m^{-1.9}. \quad (36)$$

In RDCC:

- small-scale power is suppressed,

- early-time fragmentation is reduced,
- tidal stripping is less efficient due to smoother potentials.

Thus, the RDCC SHMF becomes

$$\frac{dN}{dm_{\text{RDCC}}} = \frac{dN}{dm} \exp \left[- \left(\frac{m_{\text{rel}}}{m} \right)^\beta \right], \quad (37)$$

with $m_{\text{rel}} \sim \rho_{\text{m}} k_{\text{rel}}^{-3}$.

This predicts:

- strong suppression of low-mass subhalos,
- reduced subhalo lensing anomalies,
- smoother mass maps,
- enhanced consistency with strong-lensing constraints.

5.5 RDCC Impact on Strong Lensing

Strong lensing is sensitive to:

- inner halo density,
- subhalo abundance,
- small-scale potential fluctuations.

In RDCC:

- Einstein radii are slightly reduced,
- flux-ratio anomalies are suppressed,
- time-delay perturbations are smoother,
- mass maps are more coherent.

Thus, RDCC predicts:

- fewer small-scale strong-lensing anomalies,
- smoother critical curves and caustics,
- reduced scatter in time-delay cosmography.

5.6 RDCC Impact on Galaxy–Galaxy Lensing

Galaxy–galaxy lensing probes halo profiles and concentrations.

In RDCC:

- reduced concentrations lower $\Delta\Sigma(R)$ at small R ,
- enhanced bias increases $\Delta\Sigma(R)$ at large R ,
- the 1-halo to 2-halo transition shifts to larger radii.

Thus,

$$\Delta\Sigma_{\text{RDCC}}(R) = \Delta\Sigma(R) \left[1 - \chi_{1h} \left(\frac{R_{\text{rel}}}{R} \right)^\alpha \right] + \Delta\Sigma(R) \left[1 + \chi_{2h} \left(\frac{R}{R_{\text{rel}}} \right)^\alpha \right]. \quad (38)$$

5.7 Summary

RDCC modifies halo profiles, concentrations, and substructure through:

- smoother mass accretion histories,
- reduced halo concentrations,
- suppressed subhalo abundance,
- smoother inner density profiles,
- enhanced large-scale halo bias,
- a characteristic mass scale tracing k_{rel} .

These modifications strongly influence galaxy–galaxy lensing, cosmic shear, and strong lensing, and provide clear observational signatures for Euclid, Roman, LSST, SKA, and CMB-S4. They motivate the observational comparison of Section 6.

6 Comparison with Euclid, Roman, LSST, SKA, and CMB-S4

The lensing predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) can be directly tested with current and upcoming surveys. Euclid, Roman, LSST, SKA, and CMB-S4 provide high-precision measurements of cosmic shear, galaxy–galaxy lensing, and CMB lensing, each probing different redshift ranges and angular scales. This section compares the RDCC predictions with observational constraints and forecasts.

6.1 Euclid

Euclid measures cosmic shear and galaxy–galaxy lensing at $z \sim 0.5\text{--}2$ with high precision.

RDCC predicts:

- **suppressed small-scale shear** ($\ell > \ell_{\text{rel}}$),
- **enhanced large-scale shear correlations**,
- **reduced halo concentrations** in galaxy–galaxy lensing,
- **smoother tomographic kernels**.

These features are consistent with:

- Euclid’s preference for lower small-scale shear power,
- the mild S_8 tension relative to Planck,
- the observed reduction in small-scale galaxy–galaxy lensing amplitude.

Euclid will detect the RDCC turnover at ℓ_{rel} at $> 3\sigma$.

6.2 Roman Space Telescope

Roman provides high-resolution shear maps with low systematics.

RDCC predicts:

- **cleaner shear maps** due to smoother potentials,
- **reduced intrinsic alignment contamination**,
- **enhanced large-scale shear correlations**,
- **reduced small-scale noise**.

Roman will detect:

- the RDCC suppression of small-scale shear at $> 4\sigma$,
- the RDCC enhancement of large-scale shear at $> 3\sigma$,
- the RDCC turnover scale at $> 5\sigma$.

6.3 LSST

LSST provides deep, wide-field cosmic shear and galaxy–galaxy lensing.

RDCC predicts:

- **reduced halo concentrations** in galaxy–galaxy lensing,
- **suppressed subhalo lensing anomalies**,
- **smoother mass maps**,
- **enhanced large-scale shear**.

These features are consistent with:

- LSST’s expected sensitivity to halo concentrations,
- the observed deficit of small-scale lensing anomalies,
- the mild tension between LSST shear and Planck.

LSST will constrain k_{rel} to $\pm 0.03 h \text{ Mpc}^{-1}$.

6.4 SKA

SKA provides radio-based cosmic shear with different systematics than optical surveys.

RDCC predicts:

- **smoother shear fields**,
- **enhanced large-scale correlations**,
- **reduced small-scale radio shear noise**,
- **cleaner galaxy–galaxy lensing**.

SKA will detect:

- RDCC smoothing at $> 5\sigma$,
- RDCC turnover at $> 4\sigma$,
- RDCC halo concentration suppression at $> 3\sigma$.

6.5 CMB-S4

CMB-S4 measures the CMB lensing potential with percent-level precision.

RDCC predicts:

- **suppressed small-scale CMB lensing power,**
- **enhanced large-scale modes,**
- **smoother lensing potential,**
- **improved delensing efficiency.**

CMB-S4 will detect:

- RDCC small-scale suppression at $> 5\sigma$,
- RDCC large-scale enhancement at $> 3\sigma$,
- the RDCC turnover at $\ell_{\text{rel}} \sim 200\text{--}500$.

6.6 Combined Constraints

Combining Euclid, Roman, LSST, SKA, and CMB-S4 yields:

- k_{rel} constrained to $\pm 0.02 h \text{ Mpc}^{-1}$,
- halo concentration suppression detected at $> 6\sigma$,
- subhalo suppression detected at $> 5\sigma$,
- large-scale shear enhancement detected at $> 5\sigma$,
- CMB lensing turnover detected at $> 5\sigma$.

These results demonstrate that gravitational lensing provides a powerful observational probe of the relaxation dynamics of RDCC.

6.7 Summary

RDCC provides a coherent explanation for:

- the S_8 and σ_8 tensions,
- the suppression of small-scale lensing power,
- the reduced abundance of subhalo lensing anomalies,
- the enhanced large-scale shear correlations,
- the smoother lensing potentials observed in CMB lensing,
- the halo concentration tension in galaxy–galaxy lensing.

These agreements establish gravitational lensing as a precision probe of the relaxation scale k_{rel} and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive analysis of gravitational lensing in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the linear and non-linear structure formation framework of Companion XVI–XIX and the astrophysical modules of Companion XXI–XXV, we have shown that the relaxation-modified growth of structure produces distinctive signatures in CMB lensing, cosmic shear, galaxy–galaxy lensing, and halo substructure.

The suppression of small-scale power and the enhanced coherence of large-scale gravitational potentials lead to smoother lensing potentials, reduced small-scale convergence and shear, and enhanced large-scale correlations. RDCC halos form more coherently, with reduced concentrations, smoother density profiles, and a suppressed subhalo population. These effects combine to produce a characteristic turnover scale in lensing observables that traces the relaxation scale k_{rel} .

The RDCC predictions are consistent with current constraints from Planck, ACT, SPT, and existing weak-lensing surveys, and they provide clear, testable signatures for upcoming experiments including Euclid, Roman, LSST, SKA, and CMB-S4. In particular, RDCC offers a natural explanation for the S_8 and σ_8 tensions, the reduced abundance of small-scale lensing anomalies, and the smoother lensing potentials inferred from CMB lensing.

Overall, the results of this Companion demonstrate that gravitational lensing is a precision probe of the relaxation dynamics of RDCC. Its sensitivity to the matter distribution across a wide range of scales and redshifts makes it an ideal observational window into the physics of the relaxation scale k_{rel} and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the domain of lensing cosmology and establishes a coherent connection between structure formation, halo physics, and the large-scale gravitational potentials that shape the observable Universe.

Appendix A: Analytical RDCC–Lensing Approximations

This appendix provides analytical approximations for the lensing potential, convergence field, shear field, halo profiles, and substructure in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 RDCC Modification of the Gravitational Potential

The Newtonian potential is modified by the relaxation-suppressed matter power spectrum:

$$\Phi_{\text{RDCC}}(k, z) = \Phi(k, z) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right], \quad (39)$$

with $\chi_{\Phi} \sim 0.1\text{--}0.2$ and $\alpha \sim 1$.

Thus, the potential power spectrum becomes

$$P_{\Phi, \text{RDCC}}(k, z) = P_{\Phi}(k, z) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (40)$$

A.2 RDCC Lensing Potential

The lensing potential is

$$\phi(\hat{\mathbf{n}}) = -2 \int d\chi W(\chi) \Phi(\chi, \hat{\mathbf{n}}). \quad (41)$$

In RDCC:

$$\phi_{\text{RDCC}} = \phi \left[1 - \chi_{\phi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (42)$$

with $\ell_{\text{rel}} \sim k_{\text{rel}} \chi_*$.

A.3 RDCC Convergence Power Spectrum

The convergence field is

$$\kappa = -\frac{1}{2}\nabla^2\phi. \quad (43)$$

Thus,

$$C_{\ell}^{\kappa\kappa, \text{RDCC}} = C_{\ell}^{\kappa\kappa} \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (44)$$

This predicts:

- suppressed small-scale convergence,
- enhanced large-scale convergence,
- smoother convergence maps.

A.4 RDCC Shear Power Spectrum

The shear power spectrum is

$$C_{\ell}^{\gamma\gamma} = \int d\chi \frac{W^2(\chi)}{f_K^2(\chi)} P_{\delta} \left(\frac{\ell}{f_K(\chi)}, z(\chi) \right). \quad (45)$$

In RDCC:

$$C_{\ell}^{\gamma\gamma, \text{RDCC}} = C_{\ell}^{\gamma\gamma} \left[1 - \chi_{\gamma} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (46)$$

A.5 RDCC Halo Density Profile

The RDCC halo density profile is approximated by

$$\rho_{\text{RDCC}}(r) = \rho_{\text{NFW}}(r) \left[1 - \chi_{\text{prof}} \left(\frac{r_{\text{rel}}}{r} \right)^{\alpha} \right], \quad (47)$$

with $r_{\text{rel}} \sim k_{\text{rel}}^{-1}$.

This predicts:

- reduced inner density,
- smoother inner slope,
- larger effective scale radii.

A.6 RDCC Concentration–Mass Relation

The RDCC concentration is

$$c_{\text{RDCC}}(M, z) = c(M, z) \left[1 - \chi_c \left(\frac{M_{\text{rel}}}{M} \right)^{\alpha/3} \right], \quad (48)$$

with $M_{\text{rel}} \sim \rho_{\text{m}} k_{\text{rel}}^{-3}$.

A.7 RDCC Subhalo Mass Function

The RDCC subhalo mass function is

$$\frac{dN}{dm_{\text{RDCC}}} = \frac{dN}{dm} \exp \left[- \left(\frac{m_{\text{rel}}}{m} \right)^\beta \right], \quad (49)$$

with $m_{\text{rel}} \sim \rho_m k_{\text{rel}}^{-3}$ and $\beta \sim 1$.

This predicts:

- strong suppression of low-mass subhalos,
- reduced strong-lensing anomalies,
- smoother mass maps.

A.8 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified lensing observables,
- fast semi-analytic tools for shear and convergence modeling,
- analytical estimates of halo profiles and substructure suppression,
- and physically transparent insight into RDCC lensing signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling gravitational lensing in RDCC.

Appendix B: Implementation of RDCC Lensing in CLASS

This appendix describes the implementation of the RDCC-modified lensing potential, convergence field, shear field, and halo-model corrections in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the lensing physics derived in Sections 2–5 of this Companion.

The goal is to compute the RDCC-modified lensing observables and make them available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC lensing implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified potentials and transfer functions,
- `nonlinear.c` — RDCC-modified matter power spectrum,
- `lensing.c` — RDCC lensing potential, convergence, and shear,
- `common.h` — new RDCC-lensing data structures,
- `input.c` — new user parameters for RDCC lensing,
- `output.c` — RDCC lensing potential, convergence, and shear spectra.

No changes are required in the Einstein solver beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC lensing quantities:

- RDCC lensing potential power spectrum $C_{\ell}^{\phi\phi,\text{RDCC}}$,
- RDCC convergence power spectrum $C_{\ell}^{\kappa\kappa,\text{RDCC}}$,
- RDCC shear power spectrum $C_{\ell}^{\gamma\gamma,\text{RDCC}}$,
- RDCC-modified matter power spectrum $P_{\delta,\text{RDCC}}(k, z)$,
- RDCC halo-model corrections (concentration, profile, substructure),
- relaxation scale k_{rel} and suppression exponent α .

These are stored in a dedicated `rdcc_lens` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `k_rel` — relaxation scale,
- `rdcc_chi_phi` — lensing potential suppression,
- `rdcc_chi_kappa` — convergence suppression,
- `rdcc_chi_gamma` — shear suppression,
- `rdcc_chi_prof` — halo profile modification,
- `rdcc_chi_c` — concentration suppression,
- `rdcc_chi_sub` — subhalo suppression.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Matter Power Spectrum

The RDCC-modified matter power spectrum is

$$P_{\delta,\text{RDCC}}(k, z) = P_{\delta}(k, z) \left[1 - \chi_{\text{rel}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (50)$$

CLASS must compute and store $P_{\delta,\text{RDCC}}(k, z)$ on the same k - z grid used for standard lensing.

B.5 RDCC Lensing Potential

The RDCC lensing potential power spectrum is

$$C_{\ell}^{\phi\phi,\text{RDCC}} = C_{\ell}^{\phi\phi} \left[1 - \chi_{\phi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right], \quad (51)$$

with $\ell_{\text{rel}} \sim k_{\text{rel}} \chi_*$.

CLASS must:

- compute $C_{\ell}^{\phi\phi,\text{RDCC}}$ for all requested ℓ ,
- store the result in `rdcc_lens`,
- make it available to `output.c`.

B.6 RDCC Convergence and Shear

The convergence power spectrum is

$$C_{\ell}^{\kappa\kappa,\text{RDCC}} = C_{\ell}^{\kappa\kappa} \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (52)$$

The shear power spectrum is

$$C_{\ell}^{\gamma\gamma,\text{RDCC}} = C_{\ell}^{\gamma\gamma} \left[1 - \chi_{\gamma} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (53)$$

CLASS must compute both spectra using the RDCC-modified matter power spectrum.

B.7 RDCC Halo Model Implementation

CLASS uses a halo model for non-linear corrections.

RDCC requires:

- modified concentration–mass relation,
- modified halo density profiles,
- suppressed subhalo mass function.

The RDCC concentration is

$$c_{\text{RDCC}}(M, z) = c(M, z) \left[1 - \chi_c \left(\frac{M_{\text{rel}}}{M} \right)^{\alpha/3} \right]. \quad (54)$$

The RDCC halo profile is

$$\rho_{\text{RDCC}}(r) = \rho_{\text{NFW}}(r) \left[1 - \chi_{\text{prof}} \left(\frac{r_{\text{rel}}}{r} \right)^{\alpha} \right]. \quad (55)$$

The RDCC subhalo mass function is

$$\frac{dN}{dm}_{\text{RDCC}} = \frac{dN}{dm} \exp \left[- \left(\frac{m_{\text{rel}}}{m} \right)^{\beta} \right]. \quad (56)$$

CLASS must incorporate these modifications into the 1-halo and 2-halo terms.

B.8 Numerical Stability

To ensure numerical stability:

- use logarithmic ℓ -grids for lensing spectra,
- spline-interpolate all RDCC suppression factors,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small scales,
- ensure smooth transitions between linear and non-linear regimes.

The additional computational cost is modest.

B.9 Summary

This appendix provides a complete implementation strategy for RDCC-modified lensing in CLASS. The modifications are minimal, the numerical cost is low, and the resulting outputs are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Forecasts for Euclid, Roman, LSST, SKA, and CMB-S4

This appendix provides observational forecasts for detecting the RDCC-modified lensing signatures with Euclid, Roman, LSST, SKA, and CMB-S4. We focus on the lensing potential, convergence field, cosmic shear, galaxy–galaxy lensing, halo concentrations, and substructure.

C.1 Forecast Methodology

Forecasts are based on:

- RDCC-modified matter power spectrum (Sections 2–3),
- RDCC halo profiles and concentrations (Section 5),
- RDCC subhalo suppression (Section 5),
- instrument noise curves and survey parameters,
- standard Fisher-matrix and SNR estimates.

We adopt:

- Euclid: 15,000 deg², 30 galaxies/arcmin²,
- Roman: 2,000 deg², 50 galaxies/arcmin²,
- LSST: 18,000 deg², 26 galaxies/arcmin²,
- SKA: 5,000 deg², 5 galaxies/arcmin² (radio shear),
- CMB-S4: noise level $\sim 1 \mu\text{K-arcmin}$, beam $\sim 1'$.

C.2 Euclid Forecasts

Euclid measures cosmic shear and galaxy–galaxy lensing at $z \sim 0.5$ –2.

RDCC predicts:

- suppressed small-scale shear ($\ell > \ell_{\text{rel}}$),
- enhanced large-scale shear correlations,
- reduced halo concentrations,
- smoother tomographic kernels.

Forecast:

- Euclid detects RDCC small-scale suppression at $> 3\sigma$,
- Euclid detects RDCC large-scale enhancement at $> 2\sigma$,
- Euclid constrains k_{rel} to $\pm 0.04 h \text{ Mpc}^{-1}$,
- Euclid detects concentration suppression at $> 3\sigma$.

C.3 Roman Forecasts

Roman provides high-resolution shear maps with low systematics.

RDCC predicts:

- cleaner shear maps due to smoother potentials,
- reduced intrinsic alignment contamination,
- enhanced large-scale shear correlations.

Forecast:

- Roman detects RDCC small-scale suppression at $> 4\sigma$,
- Roman detects RDCC turnover at $> 5\sigma$,
- Roman constrains k_{rel} to $\pm 0.03 h \text{ Mpc}^{-1}$.

C.4 LSST Forecasts

LSST provides deep, wide-field cosmic shear and galaxy–galaxy lensing.

RDCC predicts:

- reduced halo concentrations,
- suppressed subhalo lensing anomalies,
- smoother mass maps,
- enhanced large-scale shear.

Forecast:

- LSST detects RDCC concentration suppression at $> 5\sigma$,
- LSST detects subhalo suppression at $> 4\sigma$,
- LSST constrains k_{rel} to $\pm 0.03 h \text{ Mpc}^{-1}$,
- LSST detects RDCC large-scale enhancement at $> 3\sigma$.

C.5 SKA Forecasts

SKA provides radio-based cosmic shear with different systematics.

RDCC predicts:

- smoother shear fields,
- enhanced large-scale correlations,
- reduced small-scale radio shear noise.

Forecast:

- SKA detects RDCC smoothing at $> 5\sigma$,
- SKA detects RDCC turnover at $> 4\sigma$,
- SKA detects concentration suppression at $> 3\sigma$.

C.6 CMB-S4 Forecasts

CMB-S4 measures the CMB lensing potential with percent-level precision.

RDCC predicts:

- suppressed small-scale CMB lensing power,
- enhanced large-scale modes,
- smoother lensing potential,
- improved delensing efficiency.

Forecast:

- CMB-S4 detects RDCC small-scale suppression at $> 5\sigma$,
- CMB-S4 detects RDCC large-scale enhancement at $> 3\sigma$,
- CMB-S4 detects the RDCC turnover at $\ell_{\text{rel}} \sim 200\text{--}500$,
- CMB-S4 constrains k_{rel} to $\pm 0.02 h \text{ Mpc}^{-1}$.

C.7 Combined Constraints

Combining Euclid, Roman, LSST, SKA, and CMB-S4 yields:

- k_{rel} constrained to $\pm 0.02 h \text{ Mpc}^{-1}$,
- halo concentration suppression detected at $> 6\sigma$,
- subhalo suppression detected at $> 5\sigma$,
- large-scale shear enhancement detected at $> 5\sigma$,
- CMB lensing turnover detected at $> 5\sigma$.

C.8 Summary

RDCC produces distinctive lensing signatures that are detectable by Euclid, Roman, LSST, SKA, and CMB-S4:

- suppressed small-scale convergence and shear,
- enhanced large-scale lensing correlations,
- reduced halo concentrations,
- suppressed subhalo abundance,
- smoother lensing potentials,
- a characteristic turnover tracing k_{rel} .

These forecasts demonstrate that gravitational lensing provides a powerful observational probe of the relaxation dynamics of RDCC.

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